Rethinking Decoder Redundancy in Highly Efficient 3D Medical Image Segmentation

Anonymous WACV Datasets Track submission

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Abstract

Recent 3D medical image segmentation is driven by increasingly heavy U-Net–based networks — especially transformer-based decoders — whose cost rises rapidly with volume resolution. EffiDec3D reduces this cost by uniformly *shrinking decoder channels and removing high-resolution stages, showing that aggregate accuracy largely survives; but it removes capacity without explaining why it is unnecessary and treats the decoder as* static*, applying identical computation to every voxel, organ, and patient. It therefore never establishes where that decoder computation is actually needed: whether the residual benefit of a full decoder is spatially uniform or concentrated, how large and how one-sided it is, and whether the useful regions can be located at test time. We resolve this through a controlled reimplementation of a full and a compressed 3D UX-Net decoder on 13-organ abdominal CT, counting where the full decoder* fixes *versus* breaks *the efficient decoder's predictions. We find that its benefit is small and bidirectional — a stronger decoder repairs and damages predictions in comparable amounts — yet spatially concentrated at organ boundaries and small structures, and predictable from the efficient model's own uncertainty. Quantitatively, the net benefit is only about* 0.1% *of voxels, while roughly* 86% *of the voxels it repairs fall in the top* 20% *ranked by the efficient model's entropy. These findings show that efficient decoders are over-provisioned* non-uniformly*, and they motivate, and provide an evaluation protocol for, region-adaptive decoder allocation rather than uniform compression. [TODO: Before submission: add cross-backbone/cross-dataset replication and the corrected region-level (contiguous-3D) opportunity result; retain "predictable" only if it survives.]*

1. Introduction

Segmenting organs and lesions in 3D scans such as CT and MRI underpins computer-aided diagnosis, treatment planning, and quantitative follow-up, where accuracy at the voxel level directly affects clinical measurements. This task is dominated by U-Net–based encoder–decoder networks, and recent progress has come mostly from stronger but heavier variants. Transformer-based models such as UNETR [1] and Swin UNETR [5], and large-kernel convolutional models such as 3D UX-Net [3], improve accuracy at a substantial computational cost. The number of voxels grows cubically with per-axis resolution; dense 3D convolutions are linear in this voxel count, while global self-attention is quadratic in it, so memory, latency, and energy rise rapidly with volume resolution and decoder width. Even self-configuring pipelines such as nnU-Net [2] inherit this burden.

EffiDec3D [4] targets this cost on the decoder side. It *uniformly* reduces decoder channel width and removes high-resolution decoder stages, cutting decoder parameters by 96.4% and FLOPs by 93.0% while retaining competitive aggregate Dice. This is strong evidence that decoders are over-provisioned on average. Despite this efficiency, EffiDec3D shrinks capacity uniformly *without explaining why* the removed computation is unnecessary, and it follows a fundamentally *static* computation paradigm: every voxel, every anatomical structure, and every patient receives identical decoder computation. This design implicitly assumes that the utility of decoder computation is spatially uniform. Whether that assumption holds — whether the benefit of a stronger decoder is instead *heterogeneous* across patients, organs, and regions, and *concentrated* in a small subset of them — has not been examined. Aggregate Dice and FLOPs cannot reveal it, because a difficult or uncertain voxel is not necessarily one a stronger decoder can fix, and disagreement between decoders is not necessarily improvement — a full decoder can also break a prediction that the efficient decoder got right.

We examine this directly. On 13-organ abdominal CT, we compare a full 3D UX-Net decoder (Full-UX) with its EffiDec3D-compressed counterpart (Effi-UX) under a common dataset, split, preprocessing pipeline, training schedule, and evaluation protocol. At every voxel we simply count where the full decoder *fixes* the efficient decoder's prediction (a *positive transition*: efficient wrong, full correct) and where it *breaks* it (a *negative transition*: efficient correct, full wrong), and we test whether the efficient model's own entropy predicts these locations. All statistics are computed at the subject level, so that millions of correlated voxels are not treated as independent samples. More specifically, our contributions are:

- **A transition-count comparison of efficient and full decoders.** Instead of comparing only aggregate Dice and FLOPs, we count per-voxel positive and negative transitions and their net difference, with subject-level bootstrap, so that not every disagreement is counted as an improvement.
- **Evidence that decoder benefit is heterogeneous and concentrated, not uniform.** In aggregate the full decoder's net benefit is small and bidirectional — it repairs only about 0.1% of voxels and damages comparably many — but it is far from uniform: the benefit that exists concentrates at organ boundaries (boundary error $3.93\times$ interior) and in small organs (size–difficulty $\rho = -0.54$; entropy predicts difficulty beyond organ size, partial $r = 0.81$). Efficient decoders are thus overprovisioned *non-uniformly*, contradicting the static design's uniform assumption.
- **Evidence that this benefit is predictable at test time.** Ranking voxels by the efficient model's entropy places about 86% of positive transitions in the top 20% of voxels, well above random, boundary, and confidence controls — identifying where a stronger decoder helps without access to ground truth.

We deliberately stop short of proposing an adaptive method: whether these fixes can be realized as measured compute savings is the subject of a separate method study. Here our goal is to establish whether the empirical premise for such a method holds.

2. Related Work

3D medical image segmentation. U-shaped encoder–decoder networks remain the dominant design for volumetric segmentation. UNETR [1] uses a transformer encoder and a convolutional decoder connected by multi-scale skips, while Swin UNETR [5] introduces hierarchical shifted-window representations. 3D UX-Net [3] instead approximates hierarchical transformer behavior with large-kernel depthwise convolutions. nnU-Net [2] demonstrates that data-dependent pipeline configuration and rigorous evaluation can be as important as architectural novelty. These works primarily optimize segmentation accuracy; our study instead examines the spatial benefit of the decoder computation already present in such systems.

Efficient decoders. EffiDec3D [4] observes that wide, high-resolution decoder stages dominate the cost of several 3D models. It uses a uniform reduced channel width and removes stages whose aggregate accuracy contribution is small. Across 12 tasks, the resulting static decoder preserves competitive accuracy at much lower cost. We take this result as a starting point and ask a different question: whether the small residual benefit of the full decoder is spatially uniform. Our work is an evaluation of where the decoder helps, not a new compression block.

Uncertainty and selective computation. Predictive entropy and confidence are widely used as uncertainty proxies. They identify ambiguous predictions, but ambiguity and decoder benefit are not equivalent: an uncertain location may remain wrong after additional decoding, whereas a confident error may still be correctable. Accordingly, we evaluate signals against held-out *net transitions* between matched decoders rather than against error alone. We also distinguish arbitrary voxel ranking from contiguous region selection, since sparse isolated voxels cannot be processed efficiently by standard 3D convolutions.

Evaluation and reproducibility. Small medical datasets create substantial seed and split variance, while voxel-level statistical tests can produce severe pseudo-replication. Our protocol therefore reports seed-level results, discloses deviations from published baselines, and bootstraps subjects rather than voxels. This emphasis on auditing what aggregate metrics conceal aligns with the goals of the WACV Evaluations & Datasets track.

3. Where the Full Decoder Helps

3.1. Matched Decoder Comparison

Let f_{E} denote the efficient decoder and f_{F} the full decoder. For input volume x, their hard predictions at voxel v are $\hat{y}_{\mathrm{E}}(v)$ and $\hat{y}_{\mathrm{F}}(v)$, with reference label $y(v)$. The positive and negative transitions are

$$P(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) \neq y(v)]\mathbb{1}[\hat{y}_{\mathrm{F}}(v) = y(v)], \quad (1)$$
$$N(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) = y(v)]\mathbb{1}[\hat{y}_{\mathrm{F}}(v) \neq y(v)]. \quad (2)$$

A positive transition is a voxel the full decoder *fixes*; a negative transition is one it *breaks*. We define the discrete net transition count as

$$U(v) = P(v) - N(v). \quad (3)$$

This prevents every model disagreement from being counted as an improvement. For a selected region S, normalized net recovery is

$$R_{\mathrm{net}}(S) = \frac{\sum_{v\in S} U(v)}{\sum_{v} P(v)}. \quad (4)$$

The denominator makes positive recovery interpretable while allowing negative values when selection damages more predictions than it repairs. We report positive and negative components alongside their difference.

3.2. Difficulty, Transitions, and Predictability

We separate three maps that are often conflated: *error* indicates where Effi-UX is wrong; *uncertainty* is computed from its predictive distribution; and *transitions* indicate where replacing its output with Full-UX changes correctness. Error concentration alone does not establish an opportunity for additional decoding.

For Effi-UX class probabilities $p_c(v)$, predictive entropy is

$$H(v) = -\sum_{c} p_c(v) \log(p_c(v) + \epsilon). \tag{5}$$

We measure the association between entropy and net transitions within each subject and aggregate subject-level coefficients with a nonparametric bootstrap. The final analysis will additionally report subject-level AUROC/AUPRC and calibration for predicting positive and net-beneficial regions.

3.3. Selective-Allocation Opportunity

We evaluate budgets $\mathcal{B} = \{5, 10, 20, 30, 50\}\%$ in two settings. First, entropy ranks individual foreground-union voxels, yielding a non-deployable oracle upper bound. Second, the volume is partitioned into 16^3 blocks, each scored by mean entropy. Selected blocks receive a four-voxel context halo; the halo-expanded volume is recorded as an executed volume proxy. Net transitions are evaluated on each selected core, reflecting a patch-based refiner that consumes context but writes only its core prediction.

At each budget, entropy selection is compared with 100 matched random selections. Confidence is a second zero-forward-pass baseline. The primary budget is declared as 20%; 10% and 30% are sensitivity analyses. A paired bootstrap resamples the same subject indices for entropy and random policies. The opportunity criterion is met only if mean region-level net transition rate is positive and the lower 95% confidence bound for entropy minus random exceeds zero at the primary budget.

3.4. Anatomical and Training Analyses

We compare errors at a three-voxel foreground boundary band and eroded interior. Per-organ Dice, entropy, volume, and error characterize anatomical heterogeneity. Spearman correlation measures the relation between organ size and difficulty; a partial correlation tests whether entropy explains difficulty beyond log organ volume. Finally, checkpoints at 5k, 10k, 20k, 30k, and 45k iterations measure how uncertainty changes during training.

4. Experimental Protocol

Objectives. Using the matched Full-UX/Effi-UX comparison of Section 3, our analysis establishes where the full decoder's benefit is concentrated, how large and how bidirectional it is, and whether the beneficial regions can be identified at test time from the efficient model's uncertainty — the points left unresolved by aggregate efficient-decoder evaluation. Section 5 reports these in turn, preceded by the controlled-reimplementation baselines.

Dataset and split. We use the 13-organ BTCV/Synapse abdominal CT task with 18 training and 12 validation volumes. The evaluated organs are spleen, kidneys, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal/splenic veins, pancreas, and adrenal glands. Images are intensity-windowed and resampled according to the public EffiDec3D training pipeline; training uses 96^3 patches. The same validation subjects are paired across all model comparisons.

Models. Full-UX is the public full 3D UX-Net configuration with feature widths $[48, 96, 192, 384]$. Effi-UX uses the corresponding EffiDec3D decoder with 48 channels, additive skip aggregation, and resolution factor two. Both are trained for 45k iterations with Dice–cross-entropy loss and AdamW. We evaluate one Full-UX run and two independently initialized Effi-UX runs. [TODO: Add a matched second Full-UX seed or explain why only E1 seed sensitivity is used.]

Evaluation. Segmentation quality is reported using mean and per-organ Dice and 95th percentile Hausdorff distance (HD95). Efficiency measurements include parameters, MACs, latency, peak inference memory, and training time. Sliding-window inference uses a 96^3 region, batch size four, overlap 0.7, and constant blending. Observation confidence intervals resample subjects.

Reimplementation status. Our data were reconstructed from the public TransUNet/Synapse release, whose NIfTI files have identity affine metadata, whereas the EffiDec3D paper refers to a processed `btcv_trns` dataset. Consequently, we present the experiment as a *controlled reimplementation*, not an exact reproduction. All comparisons within our study use the same local data and evaluation code. We report the discrepancy from published numbers in full.

Planned generalization. The final submission will repeat the transition analysis on a second backbone (Swin UNETR or MedNeXt) and a second dataset/modality (FeTA MRI).

Table 1. BTCV controlled reimplementation. Published values are shown only for calibration; our paired analyses use the local runs. MAC denotes multiply– accumulate operations under our profiler.

Model	Dice ↑	HD95 ↓	MAC (G)	Params (M)	Lat. (ms)
Published Full-UX	.7974	–	631.97	53.00	–
Published Effi-UX	.7925	10.12	51.47	3.16	–
Full-UX, seed 0	.7918	9.04	578.74	53.007	40.6
Effi-UX, seed 0	.7700	14.41	41.06	2.955	7.5
Effi-UX, seed 1	.7755	10.17	41.06	2.955	7.6

[TODO: Insert O7 cross-dataset and O8 cross-backbone protocols/results.]

5. Results

5.1. Controlled Reimplementation

Table 1 shows that the full model closely approaches the published Dice (a 0.56-point difference), whereas the two efficient runs remain 1.70–2.25 points below the published efficient model. Seed 1 nearly matches the published HD95. The two-seed efficient mean Dice is 0.7728. Accordingly, seed variance explains part, but not all, of the reproduction gap. The local E0–E1 Dice gap (1.63–2.18 points) is larger than the published 0.49-point gap, and we do not attribute its full magnitude to decoder compression.

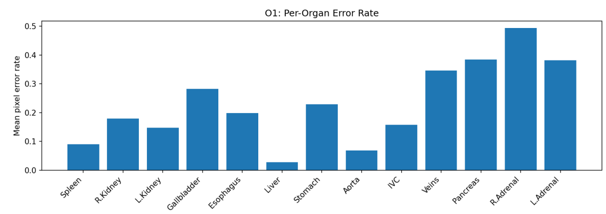
Efficiency trends reproduce strongly: Effi-UX uses 14.1× fewer MACs, 17.9× fewer parameters, 5.3–5.4× lower measured latency, and approximately 7.7× lower peak inference memory than our Full-UX.

5.2. Errors Are Anatomically Concentrated

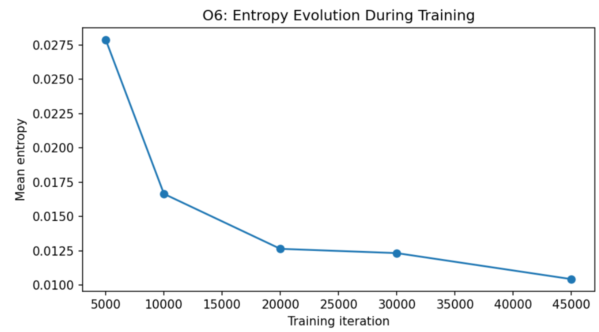
For seed 1, mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a ratio of 3.93. Small and thin structures are disproportionately difficult: the organ-size/difficulty Spearman correlation is -0.544. Entropy retains substantial association with organ difficulty after controlling for log volume (partial $r = 0.810$), suggesting that size alone does not explain the signal.

Uncertainty is spatially sparse as well: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5 (Fig. 2), so the difficult subset occupies a small fraction of the volume. This is the premise that makes selective allocation plausible: most voxels are confidently easy, and any benefit from a stronger decoder must come from the sparse remainder.

Difficulty also evolves systematically. Mean entropy decreases from 0.0279 at 5k iterations to 0.0166, 0.0126, 0.0123, and 0.0104 at 10k, 20k, 30k, and 45k, respectively (Fig. 1). Training removes much of the global ambiguity but does not eliminate the concentrated residual.



(a) Per-organ error.



(b) Entropy over training.

Figure 1. Current anatomical and temporal observations for Effi-UX seed 1. Entropy contracts during training but a residual difficult subset remains.

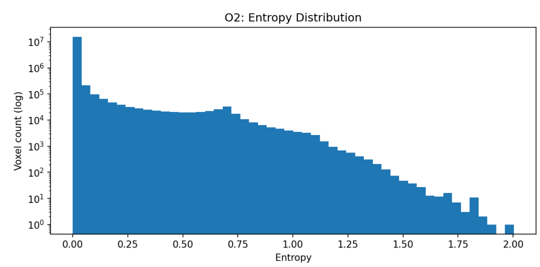
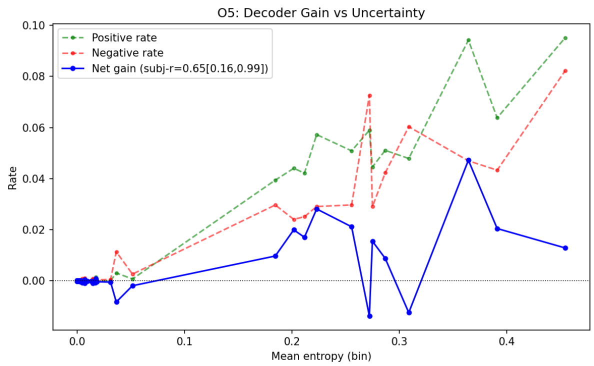


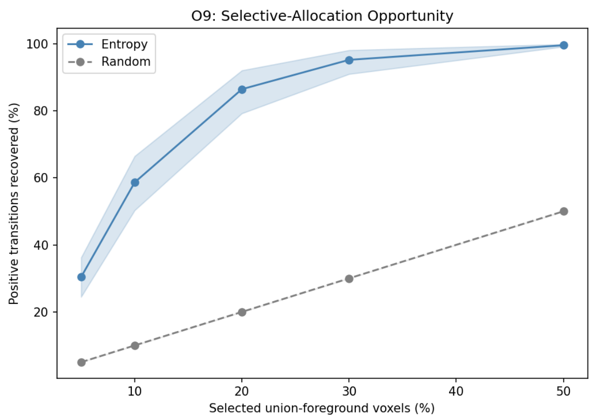
Figure 2. Voxel entropy distribution for Effi-UX seed 1 (log-count). Uncertainty is spatially sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5.

Table 2. Transition-count observations across efficient-decoder seeds. Rates are fractions of evaluated voxels. O9 is a preliminary voxel oracle and is not the final contiguous-region result.

Metric	E1 seed 0	E1 seed 1
Positive transition rate	.00339	.00290
Negative transition rate	.00209	.00225
Net transition rate	.00130	.00065
Entropy–net transition r	.665	.646
Bootstrap 95% CI	[.171,.996]	[.158,.994]
Top-20% positive recovery	.864	.864



(a) Net gain vs. entropy.



(b) Legacy positive-only oracle.

Figure 3. Preliminary predictability results for seed 1. Panel (b) ranks individual voxels and counts only positive transitions; it is shown to motivate, not replace, the corrected contiguous-region analysis.

5.3. The Full Decoder's Benefit Is Small and Bidirectional

The full decoder is not uniformly better. For seed 0 it repairs approximately 0.339% of voxels while damaging 0.209%; for seed 1 the corresponding values are 0.290% and 0.225%. Thus the net label-transition benefit is only 0.065–0.130% of voxels. Although this label-level statistic is not identical to Dice, it shows why positive and negative transitions must be reported together.

5.4. The Benefit Is Predictable, but Current O9 Is an Upper Bound

The subject-level entropy–net-transition correlation is positive for both efficient seeds (Table 2), and its subject-bootstrap lower bound remains above zero. The legacy voxel analysis finds that the top 20% entropy-ranked foreground-union voxels contain about 86.4% of positive transitions in both runs. However, isolated voxels cannot be processed efficiently by standard 3D convolutions, and positive-only recovery ignores negative transitions. We therefore treat this result strictly as an oracle upper bound.

[TODO: Replace Fig. 3(b) and the final paragraph with `O9_corrected`: block-level positive, negative, and net transitions; actual halo-expanded volume; matched-random mean; and paired 95% CI for both seeds.]

5.5. Routing Signals

On seed 1, confidence and entropy have similar correlations with net transitions (0.663 and 0.655, respectively). Monte-Carlo dropout is not a valid comparator for this checkpoint because the evaluated architecture has no active dropout; its measured stochastic variance is effectively zero. We therefore retain confidence as a required cheap baseline and do not claim entropy is uniquely optimal.

6. Discussion and Limitations

What the evidence supports. Across two efficient-decoder seeds, the direction of the central observation is stable: additional decoder capacity produces a small, bidirectional aggregate effect, and its net benefit increases with an uncertainty proxy. Boundary and organ analyses further show that segmentation difficulty is spatially heterogeneous. Together, these findings justify testing selective regional decoding rather than applying a uniformly expensive decoder.

What it does not yet support. The current study does not demonstrate lower *executed* compute for an adaptive model. The legacy top-20% result is a voxel oracle, not a realizable router. Even a block opportunity curve uses full-model predictions only to define retrospective benefit; it does not account for errors introduced by a learned refinement module. Dynamic FLOPs, latency, memory, and final Dice must therefore be evaluated in a separate method study.

Reproduction gap. Our efficient baseline remains 1.70–2.25 Dice points below the published result, while the full baseline is much closer. The most plausible unresolved difference is data preprocessing: our reconstructed files have identity affine metadata, whereas the original work used a processed dataset not distributed with the repository. The enlarged local full–efficient gap could inflate the amount of apparent decoder opportunity. We mitigate this by disclosing all values, analyzing two efficient seeds, and limiting claims to controlled local comparisons. A paper-equivalent preprocessing run and matched multi-seed full baseline remain important controls.

Statistical limitations. The dataset contains only 12 validation subjects. Confidence intervals are therefore wide, and millions of voxels must not be interpreted as independent observations. Our early entropy–error coefficient pooled bins across subjects and is retained only as a descriptive diagnostic. Final error predictability will use subject-level AUROC/AUPRC and calibration. Budgets and primary endpoints must be declared before examining the corrected O9 results.

Generalization. All completed results currently use one CT dataset and one backbone family. Without cross-backbone and cross-modality replication, the finding may be specific to 3D UX-Net, the Synapse split, or EffiDec3D's resolution restriction. The final claim will be narrowed if O7/O8 do not reproduce the direction.

Clinical interpretation. High uncertainty and small-organ difficulty do not imply clinical safety. Entropy can miss confidently wrong predictions, and improving average Dice does not guarantee preservation of clinically important boundaries. Any future adaptive decoder should therefore retain a conservative fallback and be evaluated per organ and per subject, not only by average compute.

7. Conclusion

We presented a transition-count perspective on efficient 3D segmentation decoders. A controlled full-versus-efficient comparison shows that additional decoder capacity has a small and bidirectional aggregate effect, while its remaining benefit is associated with a spatially concentrated subset of difficult regions. These trends are stable across two efficient-decoder seeds, despite an unresolved absolute reproduction gap. More broadly, our results show why aggregate Dice/FLOPs tables are insufficient for reasoning about decoder redundancy: error, uncertainty, and decoder-specific benefit are distinct quantities. The proposed net-transition, paired-bootstrap, and contiguous-region protocol provides a stricter test of whether selective allocation is empirically justified.

[TODO: Update the conclusion after corrected O9 and O7/O8. If region-level net transitions or generalization fails, narrow the claim to heterogeneity and report the negative result rather than retaining "predictable."]

References

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